

Traffic Sign Prediction Report

Anonymous

1. Introduction

This study addresses the traffic sign classification problem using the German Traffic Sign Recognition (GTSRB) dataset, which distributes traffic signs into 43 categories. The aim is to explore various classification approaches and determine the most appropriate one for the given task. This Study explores deep learning based features using ResNet18 to build on the feature set comprising pre-extracted computer vision features such as HOG features, colour histograms, etc.

Class imbalance within the dataset was explored and addressed through Synthetic Minority Oversampling (SMOTE) and class weighting.

Findings show that SVM with SMOTE and feature selection achieved the best performance among all other approaches explored in this study.

This study also analyses potential causes underlying the performance differences of the approaches explored, which includes examining feature importance and individual model effectiveness on high dimensionality.

2. Methodology

2.1 Dataset and Preprocessing

The dataset used in this study consists of 5488 traffic sign images distributes across 43 traffic sign classes/categories. The dataset appears to have class imbalance, with some classes containing less than 50 instances while others contain over 300 (Figure 1).

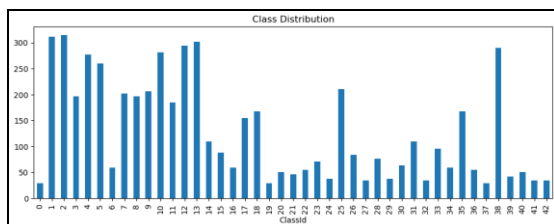


Figure 1- Bar chart showing the class distribution of the training dataset.

The dataset includes three groups of

pre-extracted features. Firstly, **colour histograms**, which are the binned representation of colour distributions in the traffic sign images. Secondly the **histogram of oriented gradients (HOG)**, composing of features extracted from grayscale images and then reduced using **Principal Component Analysis (PCA)**. Thirdly the additional features which include edge density, texture variance, and average colour channels.

These features will be used as the base of the classification approach. However, to form a more meaningful training dataset to enhance model performance, feature engineering techniques were implemented.

2.2 Feature Engineering

To extend on the features included in the training dataset, a pre-trained **ResNet18** convolutional neural network was used. This approach relied on the hierarchical representations of visual patterns learned by the deep learning model, originally trained on the **ImageNet** dataset, that is generally transferrable to other image classification tasks.

2.2.1 CNN Adaptation

The initial ResNet18 was modified by removing the **final fully connected classification layer**, creating a feature extractor that outputs a 512-dimensional feature vector for each image.

2.2.2 Image Preprocessing

ResNet18 required a certain input format. The traffic sign images were resized to 224x224, converted to RGB format, and normalized using ImageNet's means and standard deviations to match the expected data distribution.

2.2.3 Feature Extraction and Addition

The pre-processed images are then passed into the modified ResNet18 model to produce a 512-dimensional feature vector for each image.

The features extracted by the CNN was then

merged with the training dataset including the colour histogram features, HOG features, and additional features.

2.3 Feature Selection

As Random Forest can determine feature importance, the classifier was trained on the training dataset to leverage this function. This study explored different feature selection sizes such as the top 100, 200, 300 and 400, with 200 showing the most significant improvement to the dataset.

2.4 Train-Validation Split

A stratified train-validation split was implemented using an 80-20 ratio while also preserving the initial class proportions of the 43 traffic sign categories. This approach ensures a fair comparison across models.

2.5 Model Implementation

In this study, three different machine learning models were explored.

2.5.1 Support Vector Machine (SVM)

The SVM classifier was configured with a **radial basis function (RBF)** kernel. This allowed the model to find non-linear decision boundaries in the feature space, which made sense for the current training dataset with high dimensionality.

2.5.2 Random Forest

The Random Forest implementation for this study uses 100 decision trees with unrestricted depth, which leverages from the multiple decision trees constructed during training and outputs the class that is the mode of the individual tree predictions.

2.5.3 K-Nearest Neighbours (KNN)

KNN was implemented with $k = 3$. Multiple k values such as 2, 3, 4, 5, 6 were explored, with $k = 3$ showing the most accurate outputs.

2.5.4 Class Weighting and Minority Oversampling

To address the class imbalance in the training dataset, two approaches were explored.

First, **class weights** were applied inversely proportional to the class frequencies in the training data. In other words, minority classes were assigned higher weights.

Second, the **Synthetic Minority Oversampling Technique** was implemented to create synthetic instances for minority classes by interpolating between existing

instances of the minority classes.

3 Results

3.1 Evaluation Metrics

To get a broad and comprehensive evaluation of the performance of each classification model/approach, multiple performance metrics were used. This includes **accuracy, precision recall, and F1-score**.

These metrics are all useful in the case of traffic safety, as accuracy shows the general performance, high precision avoids false alarms, and high recall avoids potentially missing crucial traffic signs.

3.2 Model Performance

Model	Accuracy	Macro Avg Precision
Base SVM	93%	0.96
Random Forest	79%	0.83
KNN	85%	0.82
SVM with Class Weights	93%	0.96
SVM with SMOTE	93%	0.96
SVM with Feature Selection	94%	0.95
SVM with Weights & FS	93%	0.95
SVM with SMOTE & FS	94%	0.95

Table 1- Shows the accuracy and average precision across all classes for each model approach.

Model	Macro Avg Recall	Macro Avg F1	Weighted Avg F1
Base SVM	0.89	0.92	0.93
Random Forest	0.66	0.70	0.78
KNN	0.88	0.84	0.86
SVM with Class Weights	0.89	0.92	0.93
SVM with SMOTE	0.89	0.92	0.93
SVM with Feature Selection	0.91	0.92	0.93
SVM with Weights & FS	0.90	0.92	0.93
SVM with SMOTE & FS	0.91	0.92	0.94

Table 2- Shows the average recall, F1-score, and weighted F1-Score across all classes for each model approach.

3.2.1 SVM Performance

The results demonstrate that SVM achieved the highest accuracy (93%), significantly outperforming both Random Forest (79%) and KNN (85%). Thus, rather than exhaustively exploring all models further, this study focuses on the most promising model (SVM) for fine tuning.

3.2.2 SVM with Feature Selection

SVM with feature selection showed a slight improvement in accuracy (from 93% to 94%), and an improved average recall (from 89% to 91%).

3.2.3 Class Imbalance Techniques

Both SMOTE and class weightings achieved a similar overall performance with SVM. However, SMOTE achieved the highest overall performance among the two class imbalance techniques when coupled with feature selection (94% accuracy and 0.94 weighted F1-score) (Table 2).

3.2.4 Individual Class Performance

Class	Precision	Recall	F1-Score
0	1.00	0.50	0.67
1	0.90	0.97	0.93
2	0.85	0.87	0.86
3	0.86	0.79	0.83
4	0.90	0.98	0.94
5	0.83	0.87	0.85
6	1.00	1.00	1.00
7	0.97	0.85	0.91
8	0.94	0.87	0.91
9	1.00	0.98	0.99
10	0.96	0.98	0.97
11	0.90	1.00	0.95
12	1.00	1.00	1.00
13	1.00	1.00	1.00
14	1.00	1.00	1.00
15	1.00	1.00	1.00
16	1.00	1.00	1.00
17	1.00	1.00	1.00
18	0.97	0.91	0.94
19	1.00	0.50	0.67
20	0.89	0.80	0.84
21	1.00	0.78	0.88
22	0.90	0.82	0.86
23	1.00	1.00	1.00
24	1.00	0.75	0.86
25	0.85	0.93	0.89
26	1.00	0.94	0.97
27	0.86	0.86	0.86
28	0.81	0.87	0.84
29	1.00	0.71	0.83
30	1.00	0.92	0.96
31	0.79	1.00	0.88
32	1.00	1.00	1.00
33	0.95	1.00	0.97
34	1.00	1.00	1.00
35	0.97	0.97	0.97
36	0.91	0.91	0.91
37	1.00	0.83	0.91

Table 3- Shows the precision, recall, and F1-score of each class for the SVM using SMOTE and feature selection.

Looking specifically at the SVM model that uses SMOTE and feature selection, the results demonstrate that the model achieved perfect or near perfect classification scores for several classes including classes 1, 9, 10, 12, 13, etc. (Table 3). There is a high number of instances with these class labels in the training dataset (Figure 1).

Contradictorily, the model achieved relatively low scores for classes such as 0, 19, 27, etc. (Table 3). There is a low number of instances with these class labels in the training dataset (Figure 1).

3.2.5 Feature Importance

The feature selection via Random Forest identifies key features that contributed the most to classification performance.

Rank	Feature Name	Importance Score
1	hog_pca_3	0.018181
2	hog_pca_1	0.014648
3	hog_pca_5	0.012046
4	hog_pca_7	0.011688
5	hog_pca_0	0.008583
6	hog_pca_9	0.008514
7	hog_pca_2	0.007245
8	hog_pca_4	0.006829
9	hog_pca_8	0.006625
10	CNN_406	0.006050
11	hog_pca_12	0.005399
12	hog_pca_10	0.004876
13	CNN_167	0.004867
14	CNN_126	0.004779
15	CNN_482	0.004605
16	CNN_359	0.004449
17	CNN_398	0.004261
18	CNN_455	0.004202
19	CNN_329	0.003915
20	CNN_158	0.003837

Table 4- Shows the importance score of the top 20 features.

As shown, several CNN extracted features from ResNet18 were included in the top 20 features, along with some of the pre-extracted features.

3.2.6 Final Model Selection

Based on the findings, SVM coupled with SMOTE and feature selection was selected as the final model for this study. This decision was justified by several reasons. First, the superior accuracy (94%) and weighted F1-score (0.94). Second, the superior handling of minority class (when used with feature selection). Third, the improved performance when feature selection is applied, suggesting a more generalizable approach that may work better on unseen data. And finally, SVM's capabilities in high dimensional spaces, which align with our feature engineering approach (which creates a large feature set).

4 Discussion and Critical Analysis

4.1 Model Performance

As identified, SVM performed significantly better than both Random Forest and KNN. This can be explained by several factors.

4.1.1 SVM with High Dimensional Spaces

The feature set, especially after feature engineering with Random Forest, creates a high dimensional representation. SVM, specifically using the RBF Kernel, is particularly good for such spaces due to its ability to identify non-linear decision boundaries.

4.1.2 Random Forest Limitation

The poor performance of the Random Forest classifier was unexpected. Given that it combines the predictions of multiple decision trees trained on random subsets of data and features, it was expected to be robust and able to generalize well on unseen images. This underperformance may be due to the **curse of dimensionality** in tree based methods.

True decision boundaries may involve relationships between multiple features simultaneously. As each split only considers one feature at a time, a decision tree may need many levels of splits to approximate what the SVM model can capture in a single kernel operation. This is critical especially for traffic sign classification, as distinguishing between visually similar signs likely require integrating information across multiple features (potentially a lot particularly for traffic signs). In essence, the Random Forest classifier may not be as effective as SVM in capturing interactions between multiple features.

4.1.3 KNN Limitation

KNN achieved a reasonable accuracy (85%) considering its simplicity compared to the other two models, but still fell short of SVM's performance. Unlike with Random Forest, this outcome was more expected.

Similarly, KNN suffers from the **curse of dimensionality**. In high dimensional spaces, KNN suffers from the **concentration of distances** phenomenon. Meaning, the ratio of the distance to the nearest neighbour to the distance to the farthest neighbour approaches 1. This means that with the current feature space, each point may appear equidistant from any other point. Thus, KNN may fail to identify truly similar traffic signs.

Though, KNN still surprisingly performed better than Random Forest on the

validation set.

4.2 SMOTE vs. Weighting

The similar performance of both weighting and SMOTE indicates that both approaches are effective in addressing the class imbalance problem. However, it was shown that SMOTE coupled with feature selection produced slightly better results compared to weighting coupled with feature selection (94% as opposed to 93% respectively) (Table 1).

This slight improvement could be due to SMOTE generating synthetic instances that enrich the feature space for minority classes by providing more information, helping SVM better learn decision boundaries. In other words, weighting only alters the penalty for misclassification and doesn't actually add new information like SMOTE.

Additionally, this combination may reduce the curse of dimensionality as well since synthetic examples may be more impactful in a refined feature space.

4.3 Feature Analysis

4.3.1 Feature Set

The implementation of CNN features on top of the pre-extracted features evidently achieves good results. This highlights the value of having a variety of features to capture different aspects of traffic signs.

For example, CNN features extract high level information, having learned hierarchical representations from many images. HOG features capture the geometrical and structural information, while colour histograms capture distinctive colour patterns that differentiate sign categories.

4.3.2 Feature Selection

The improvement in performance with feature selection suggest that some features may have introduced noise or redundancy to the dataset.

By focusing on the most informative features, this approach may mitigate overfitting and improve generalization on unseen data.

4.4 Error Analysis

Analysing the misclassifications more closely may reveal patterns that provide insights into the potential limitations of the current approach.

For example, classes with fewer instances (Figure 1) continued to show relatively low recall despite mitigation strategies (Table 3). This indicates a persistent challenge in identifying minority classes. Meaning that while the class imbalance technique used is helpful, it does not completely resolve the issue.

5 Conclusions

In the end, this study identified SVM with SMOTE and feature selection to be the most optimal approach, while demonstrating several key findings.

First, the choice of machine learning model significantly impacts performance, especially due to high dimensionality of the feature set. SVM's superior performance compared to Random Forest and KNN reinforces the importance of the model's capabilities in high-dimensional spaces and ability to capture feature interactions.

Second, implementing features with deep learning (using ResNet18 CNN) was highly effective for this task. The use of HOG, colour histogram, along with CNN features demonstrated the effectiveness of having various features showing different aspects of signs (shape, colour, etc.).

Third, class imbalance mitigation through SMOTE coupled with feature selection enhances results by addressing both data distribution and feature quality.

Future work should focus on improving minority classes and exploring ensemble methods for traffic sign predictions with large feature sets.